part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
	3	1.1.8, 1.1.7	CN	C	2001 q1

Find the equation of the straight line which is parallel to the line with equation $2x + 3y = 5$ and which passes through the point (2, –1). **3**

Give 1 mark for each •

Illustrations for awarding each •

ans: $2x + 3y = 1$ 3 marks

- •1 ss : express in standard form
- •2 ic : interpret gradient
- •3 ic : state equation of st line

- •1 $y = -\frac{2}{3}x + \frac{5}{3}$ stated or implied by •2
- •2 $m_{line} = -\frac{2}{3}$ stated or implied by •3
- •3 $y - (-1) = -\frac{2}{3}(x - 2)$

Notes

1 •3 is only available for candidates who attempt to find or state the gradient.

2 •3 is still available even though •1 and •2 may not have been awarded.

example for note 2

$m = 2$

$y - (-1) = 2(x - 2)$ earns •3

example for note 1

$y - (-1) = 7(x - 2)$ earns no marks.

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
	3	2.1.7	CN	C	2001 q2

For what value of k does the equation $x^2 - 5x + (k+6) = 0$ have equal roots? **3**

Give 1 mark for each •

Illustrations for awarding each •

ans: $k = \frac{1}{4}$ 3 marks

- $\bullet^1$ ss : know to set disc. to zero
- $\bullet^2$ ic : substitute a, b and c into discriminant
- $\bullet^3$ pd : process equation in k

- $\bullet^1$ $b^2 - 4ac = 0$ stated or implied by $\bullet^2$
- $\bullet^2$ $(-5)^2 - 4 \times (k+6)$
- $\bullet^3$ $k = \frac{1}{4}$

Notes

1 "= 0" must occur on one of the lines in the solution to this question.

2 If the expression for the discriminant involves x, no marks are available.

3 If the phrase "discriminant = 0" appears at the start, then $\bullet^1$ can only be awarded if $\bullet^2$ is awarded.

ie $disc = 0$

$25 - 4(k+6) = 0$

$k = \frac{1}{4}$

can be awarded 3 marks.

Alternative sol

- $\bullet^1$ $x^2 - 5x + k + 6 = 0$
- $\bullet^2$ $x^2 - 5x + \left(\frac{5}{2}\right)^2 = 0$ has equal roots

$k + 6 = \frac{25}{4}$

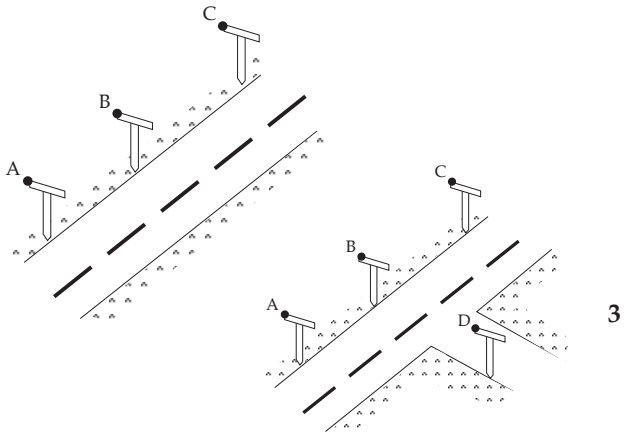
- $\bullet^3$ $k = \frac{1}{4}$

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
a)	3	3.1.7	CN	C	2001 q3
b)	3	3.1.10	CN	C	

(*a*) Roadmakers look along the tops of a set of T-rods to ensure that straight sections of road are being created. Relative to suitable axes the top left corners of the T-rods are the points A(–8, –10, –2), B (–2, –1, 1) and C(6, 11, 5). Determine whether or not the section of road ABC has been built in a straight line. **3**

(*b*) A further T-rod is placed such that D has coordinates (1, –4, 4).
Show that DB is perpendicular to AB. **3**



Give 1 mark for each •

Illustrations for awarding each •

a ans: the road ABC is straight 3 marks

- •1 ic : interpret vector (eg $\vec{AB}$)
- •2 ic: interpret multiple of vector
- •3 ic : complete proof

- •1 *e.g.* $\vec{AB} = \begin{pmatrix} 6 \\ 9 \\ 3 \end{pmatrix}$
- •2 *e.g.* $\vec{BC} = \begin{pmatrix} 8 \\ 12 \\ 4 \end{pmatrix} = \frac{4}{3}\vec{AB}$
- •3 a) a common direction exists
  and b) a common point exists
  so A, B, C collinear

Notes
1 For •3, accept references to "they" are parallel.
2 For (b) Converse of Pythagoras provides an alternative.[See below].
3 Other methods include using the cosine rule and the scalar product.

- •1 calculating $AB^2 = 126,\ BD^2 = 27,\ AD^2 = 153$
- •2 stating $\cos A\hat{B}D = \frac{126+27-153}{\sqrt{\ldots}\sqrt{\ldots}}$
- •3 $= 0$ *so* $\angle ABD = 90°$

- •1 calculating $AB^2 = 126,\ BD^2 = 27,\ AD^2 = 153$
- •2 stating $126 + 27 = 153$
- •3 by converse of Pythagoras $\angle ABD = 90°$

b ans: proof 3 marks

- •4 ic : interpret vector (ie $\vec{BD}$)
- •5 ss: state requirement for perpend.
- •6 ic : complete proof

- •4 $\vec{BD} = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix}$
- •5 $\vec{AB}.\vec{BD} = 0$
- •6 $\vec{AB}.\vec{BD} = 18 - 27 + 9 = 0$

- •4 $\vec{BD} = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix}$
- •5 $\vec{AB}.\vec{BD} = 18 - 27 + 9$
- •6 $= 0$ so AB is at right angles to BD

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
	2	1.2.8	NC	C	2001 q4

Given $f(x) = x^2 + 2x - 8$, express $f(x)$ in the form $(x+a)^2 - b$. **2**

Give 1 mark for each •

ans: $(x+1)^2 - 9$ 2 marks

- •1 ss : eg start to complete square
- •2 pd : complete process

Illustrations for awarding each •

- •1 $(x+1)^2 \ldots\ldots$
- •2 $(x+1)^2 - 9$

OR

- •1 $a = 1$
- •2 $b = 9$

OR

- •1 $x^2 + 2x - 8 \equiv x^2 + 2ax + a^2 - b$
- •2 $a = 1$ and $b = 9$

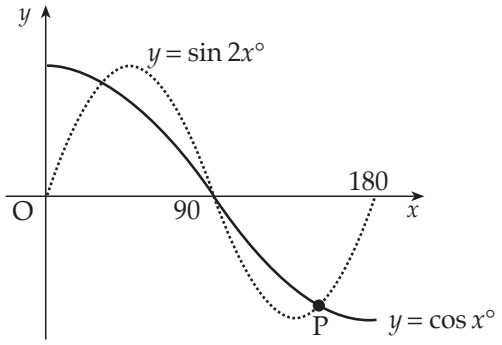
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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
	4,1	2.3.5	NC	C	2001 q5

b

5 (*a*) Solve the equation $\sin 2x° - \cos x° = 0$ in the interval $0 \le x \le 180$. **4**

(*b*) The diagram shows parts of two trigonometric graphs, $y = \sin 2x°$ and $y = \cos x°$.
Use your solutions in (*a*) to write down the coordinates of the point P. **1**



Give 1 mark for each •

Illustrations for awarding each •

a ans: 30, 90, 150 4 marks

- •1 ss : use double angle formula
- •2 pd : factorise
- •3 pd : process
- •4 pd : process

- •1 $2\sin x°\cos x°$
- •2 $\cos x°(2\sin x° - 1)$
- •3 $\cos x° = 0,\ \sin x° = \frac{1}{2}$
- •4 $90, 30, 150$

- •3 $\sin x° = \frac{1}{2}$ and $x = 30, 150$
- •4 $\cos x° = 0$ and $x = 90$

Notes

1 The inclusion of wrong answer(s) means the mark is not awarded (•4 in method 1, •3 or •4 in method 2).

b ans: $\left(150, -\frac{\sqrt{3}}{2}\right)$ 1 mark

- •5 ic : interpret graph

- •5 $\left(150, -\frac{\sqrt{3}}{2}\right)$

Notes

2 Accept $y = \cos 150° = -\frac{\sqrt{3}}{2}$ as poor form

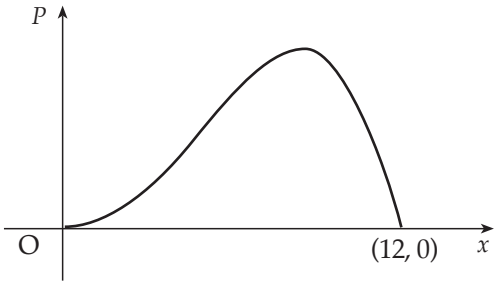
3 Wrong formula:

- •1 ✗ $2\cos^2 x° - 1$
- •2 √ $(2\cos x° + 1)(\cos x° - 1) = 0$
- •3 √ $\cos x° = -\frac{1}{2},\ \cos x° = 1$
- •4 √ $0, 120, (240, 360)$

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
	5	1.3.15	NC	C	2001 q6

A company spends x thousand pounds a year on advertising and this results in a profit of P thousand pounds. A mathematical model, illustrated in the diagram, suggests that P and x are related by $P = 12x^3 - x^4$ for $0 \le x \le 12$.
Find the value of x which gives the maximum profit.



5

Give 1 mark for each •

Illustrations for awarding each •

ans: x = 9

5 marks

•1 ss : start diff. process
•2 pd : process
•3 ss : set derivative to zero
•4 pc : process
•5 ic : interpret solutions

•1 $\frac{dP}{dx} = 36x^2 \ldots\ldots$ ***or*** $\frac{dP}{dx} = \ldots\ldots - 4x^3$
•2 $\frac{dP}{dx} = 36x^2 - 4x^3$
•3 $\frac{dP}{dx} = 0$
•4 $x = 0$ ***and*** $x = 9$
•5 nature table about $x = 9$ ***and*** $x = 9$

Notes
1 The "= 0" shown in •3 may appear anywhere in the working but must appear explicitly.

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
a)	2	1.2.6	NC	C	2001 q7
b)	5	2.3.5	NC	C	

Functions $f(x) = \sin x$, $g(x) = \cos x$ and $h(x) = x + \frac{\pi}{4}$ are defined on a suitable set of real numbers.

(*a*) Find expressions for

(i) $f(h(x))$

(ii) $g(h(x))$. **2**

(*b*) (i) Show that $f(h(x)) = \frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x$.

(ii) Find a similar expression for $g(h(x))$ and hence solve the equation $f(h(x)) - g(h(x)) = 1$ for $0 \le x \le 2\pi$. **5**

Give 1 mark for each •

a ans: $\sin\left(x + \frac{\pi}{4}\right)$, $\cos\left(x + \frac{\pi}{4}\right)$ 2 marks

•1 ic : interpret composite functions

•2 ic : interpret composite functions

Illustrations for awarding each •

•1 $\sin\left(x + \frac{\pi}{4}\right)$

•2 $\cos\left(x + \frac{\pi}{4}\right)$

Notes

1 One mark may be awarded for "$f\left(x + \frac{\pi}{4}\right)$ ***and*** $g\left(x + \frac{\pi}{4}\right)$"

2 For $\sin x + \frac{\pi}{4}$ *and* $\cos x + \frac{\pi}{4}$ award 1 mark only, unless there is evidence in part b that they have been expanded correctly, in which case treat as bad form and award 2 marks.

3 Do not penalise the appearance of 45°

b ans: proof and $x = \frac{\pi}{4}, \frac{3\pi}{4}$ 5 marks

•3 ss : expand $\sin\left(x + \frac{\pi}{4}\right)$

•4 ic : interpret

•5 ic : substitute

•6 pd : start solving process

•7 pd : process

•3 $\sin x \cos\frac{\pi}{4} + \cos x \sin\frac{\pi}{4}$ ***and complete***

•4 $g(h(x)) = \frac{1}{\sqrt{2}}\cos x - \frac{1}{\sqrt{2}}\sin x$

•5 $\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right) - \left(\frac{1}{\sqrt{2}}\cos x - \frac{1}{\sqrt{2}}\sin x\right)$

•6 $\frac{2}{\sqrt{2}}\sin x$

•7 $x = \frac{\pi}{4}, \frac{3\pi}{4}$

Notes

4 If the evidence for •5 has no brackets, •5 can only be awarded if there is evidence further on that brackets has been implied.

5 •7 is only available for answers in radians.

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
	3	3.3.3, 3.3.4	NC	C	2001 q8

Find x if $4\log_x 6 - 2\log_x 4 = 1$. **3**

Give 1 mark for each •

Illustrations for awarding each •

ans: 81 3 marks

- •1 pd : use log-to-index rule
- •2 pd : use log-to-division rule
- •3 ic : interpret base for $\log_x a = 1$ and simplify

- •1 $\log_x 6^4 - \log_x 4^2$
- •2 $\log_x \frac{6^4}{4^2}$
- •3 all processing leading to $x = 81$

Further illustrations

- •1 $4\log_x 6 - 4\log_x 2 = 1$
  $4(\log_x 6 - \log_x 2) = 1$
- •2 $4\log_x \frac{6}{2} = 1$
  $\log_x \frac{6}{2} = \frac{1}{4}$
  $x^{\frac{1}{4}} = 3$
- •3 $x = 81$

- •1 $4\log_x 6 - 4\log_x 2 = 1$
  $4(\log_x 6 - \log_x 2) = 1$
  $4\log_x \frac{6}{2} = 1$
- •2 $\log_x \left(\frac{6}{2}\right)^4 = 1$
- •3 $x = 81$

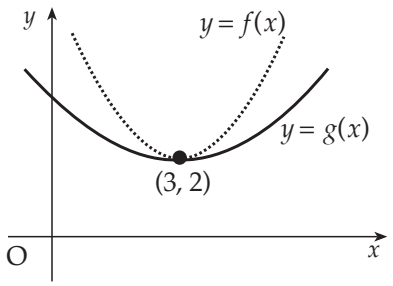
$2\log_x 6 - \log_x 4 = \frac{1}{2}$

- •1 $\log_x 6^2 - \log_x 4 = \frac{1}{2}$
- •2 $\log_x \frac{6^2}{4} = \frac{1}{2}$
  $x^{\frac{1}{2}} = 9$
- •3 $x = 81$

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
	2	1.2.4, 1.3.8	CN	C	2001 q9

The diagram shows the graphs of two quadratic functions $y = f(x)$ and $y = g(x)$. Both graphs have a minimum turning point at (3, 2).
Sketch the graph of $y = f'(x)$ and on the same diagram sketch the graph of $y = g'(x)$.



2

Give 1 mark for each •

Illustrations for awarding each •

ans: for all k 2 marks

•1 ss : use $\frac{d}{dx}(\text{"quadratic"}) = \text{"linear"}$

•2 ic : iinterpret stationary point

•1 st line for f' thr' $(3,0)$, $m_{f'} > 0$

•2 st line for g' thr' $(3,0)$, $m_{f'} > m_{g'} > 0$

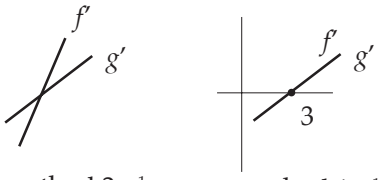
•1 st lines for f' and g ', with $m_{f'} > m_{g'} > 0$

•2 two lines intersecting at $(3,0)$

Notes

1 Award 0 marks for two curves through anywhere.

2 Further illustrations:

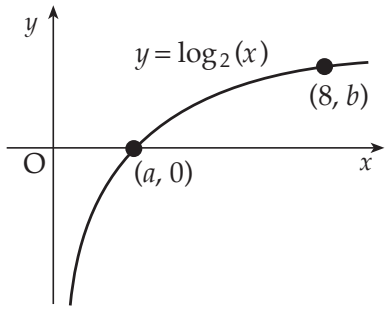


method 2 •1 method 1 •1 method 2 •2

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
a)	1	1.2.5	CN	B	2001 q10
b)	3	1.2.4, (3.0.0HG?)	CN	A	

10 The diagram shows a sketch of part of the graph of $y = \log_2(x)$.



(*a*) State the values of *a* and *b*. **1**

(*b*) Sketch the graph of $y = \log_2(x+1) - 3$. **3**

Give 1 mark for each •

a ans: $a = 1,\ b = 3$ 1 mark

•1 pd : use $\log_p q = 0 \Rightarrow q = 1$ and evaluate $\log_p p^k$.

b ans: sketch 3 marks

•2 ss : use a translation

•3 ic : identify one point

•4 ic : identify a second point

Illustrations for awarding each •

•1 $a = 1$ ***and*** $b = 3$

•2 a "log – shaped" graph of the same orientation

•3 sketch passes through $(0, -3)$ (labelled)

•4 sketch passes through $(7, 0)$ (labelled)

Notes

1 Do not penalise any errors made in relation to the asymptote, missing or otherwise.

2 $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$ is the correct translation!. You may also consider $\begin{pmatrix} -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ *and* $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$. •2 is still available and in addition one mark (from •3 an •4) may be awarded for both points consistent with the wrong translation.
Do not consider any other translation.

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part	marks	Syllabus Code	Calc. Code (CN,CR,NC)	Grade (C, B, A)	Source
a)	4	2.4.5	CN	B	2000 q11
b)	3	2.4.4	CN	B	
c)	3	2.4.4	CN	A	

Circle P has equation $x^2 + y^2 - 8x - 10y + 9 = 0$. Circle Q has centre (–2, –1) and radius $2\sqrt{2}$.

(*a*) (i) Show that the radius of circle P is $4\sqrt{2}$. **4**

(ii) Hence show that circles P and Q touch. **3**

(*b*) Find the equation of the tangent to circle Q at the point (–4, 1). **3**

(*c*) The tangent in (*b*) intersects circle P in two points. Find the *x*-coordinates of the points of intersection, expressing your answers in the form $a \pm b\sqrt{3}$.

Give 1 mark for each •

a ans: proof 4 marks

- •1 ic : interpret centre of circle (P)
- •2 ss: find radius of circle (P)
- •3 ss : find sum of radii
- •4 pd : compare with distance between centres

Illustrations for awarding each •

•1 $r_P = \sqrt{16 + 25 - 9} = \sqrt{32} = 4\sqrt{2}$

•2 $r_P + r_Q = 4\sqrt{2} + 2\sqrt{2} = 6\sqrt{2}$

•3 $C_P = (4, 5)$

•4 $C_P C_Q = \sqrt{6^2 + 6^2} = 6\sqrt{2}$ ***and*** "so touch"

• •

•1 $C_P = (4, 5)$

•2 $C_P C_Q = \sqrt{6^2 + 6^2} = \sqrt{72}$

•1 $r_P = \sqrt{16 + 25 - 9} = \sqrt{32} = 4\sqrt{2}$

•2 $r_P + r_Q = 4\sqrt{2} + 2\sqrt{2} = 6\sqrt{2} = \sqrt{72}$ ***and*** "so touch"

b ans: $y = x + 5$ 3 marks

- •5 ss : find gradient of radius
- •6 ss : use $m_1 m_2 = -1$
- •7 ic : state equation of tangent

•5 $m_r = -1$

•6 $m_{tgt} = +1$

•7 $y - 1 = 1(x + 4)$

Note

1 •7 is only available if an attempt has been made to find a perpendicular gradient.

c ans: $x = 2 \pm 2\sqrt{3}$ 3 marks

- •8 ss : substitute linear into circle
- •9 pd : express in standard form
- •10 pd : solve (quadratic) equation

•8 $x^2 + (x + 5)^2 - 8x - 10(x + 5) + 9 = 0$

•9 $2x^2 - 8x - 16 = 0$

•10 $x = 2 \pm 2\sqrt{3}$

Note

2 •8, •9 and •10 are only available when the answer to part (b) is of the form $y = ax + b$ *where* $a, b \in R,\ a \neq 0$.

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